

Software Engineering — 10-Page Master Quick Revision

High-Yield Formula Sheet, COCOMO Matrices & BIT Mesra PYQ Solutions

10-Page Master Quick Revision — Software Engineering (CS24353)

Page 1-2: Process Models (Waterfall vs. Spiral vs. Scrum), RMMM

Page 5-6: Cohesion Spectrum, Coupling Spectrum & UML Architecture

Page 9-10: Function Points, COCOMO Formulas, CMMI 5-Levels & SCM

Page 3-4: Functional/Non-Functional Requirements & IEEE 830 SRS

Page 7-8: ECP/BVA Black-Box & McCabe Cyclomatic Complexity $V(G)$

High-Yield Software Engineering Master Formulas

Metric / Concept	Exact Formula / Rule	Key Exam Insight
Cyclomatic Complexity	$V(G) = E - N + 2 = P + 1 = \text{Regions}$	Equals number of linearly independent paths.
Function Points (FP)	$FP = UFP \times [0.65 + 0.01 \times \sum F_i]$	$\sum F_i$ is sum of 14 general system characteristics.
COCOMO Effort	$E = a_b \times (KLOC)^{b_b} \text{ PM}$	Organic: $a_b = 2.4, b_b = 1.05$.
Availability	$A = \frac{MTTF}{MTTF + MTTR} \times 100\%$	$MTBF = MTTF + MTTR$.